

Efficient Adjoint-Based Shape Optimization Method for the Inverse Design of Microwave Components

Shengwei Ji^{ID}, Siteng HuYan^{ID}, Liuge Du^{ID}, Xiao Xu^{ID}, and Jia Zhao^{ID}

Abstract—An adjoint-based shape optimization approach for the inverse design of microwave components is proposed. The proposed approach is a pure postprocessing process that only needs the field solutions. It is independent on numerical methods, system matrix, and mesh information, thus can be used in conjunction with mainstream Maxwell solvers currently available on the market, which makes it an efficient, convenient, and user-friendly method. To be applicable to the shape optimization of complex cases, this approach can calculate shape derivatives at all points in design region with only one simulation by a rigorous analytical solution, which makes it have arbitrarily many degrees of freedom (DoFs) while the reported similar approaches can only deal with several variables. Flexible parameters' updating is achieved by the level set method. Owing to the speed of calculating shape derivatives at boundary points within adjoint method, coupled with the accurate updating of shape by gradient descent and level set method, our approach normally requires several or dozens of iterations even in cases with numerous DoFs. The developed approach is used to optimize a high-directivity directional coupler and an unequal Wilkinson power divider to verify its validity and robustness.

Index Terms—Adjoint method, degree of freedom (DoF), inverse design, shape derivative, shape optimization.

I. INTRODUCTION

GUIDED-WAVE devices, such as coupler and power divider, and spatial electromagnetic (EM) devices, such as antenna and spatial filter, together constitute the transmission path of EM waves in microwave systems for wireless communication and electronic countermeasures. The performance of these devices that is directly related to the manipulation of EM waves is determined by their structures, especially the shape of their metal elements. Therefore, some conventional approaches, such as the direct design method by manually changing shapes guided by engineering insight and equivalent circuit model, have been adopted to perform shape optimization [1], [2], [3], [4]. However, these methods not

only rely heavily on intuition and experience of designers but also explore only a small fraction of the available shapes. In addition, they are time-consuming and computationally intensive. More importantly, these methods can hardly achieve shape optimization fine enough to meet the increasingly precise manipulation of EM waves in the above-mentioned systems [5], [6], [7], [8].

For these reasons, a variety of methodologies have been developed in pursuit of efficient, exhaustive, and fine shape optimization of microwave components. Generally, these methodologies can be broadly classified into two types, namely, stochastic optimization and deterministic optimization. Stochastic optimization, such as particle swarm optimization, genetic optimization, or other hybrid methods tailored for specific problems, has the global searching ability endowed by the inherent randomness of search direction [9], [10], [11]. And thus, this kind of method prevails in finding the global optimal solutions. However, when stochastic method is applied for shape optimization of complex structures with numerous degrees of freedom (DoFs), the inherent randomness of search direction leads to high computational cost generated by a bulk of repeated full-wave simulations, which will prevent the realization of finding all possible designs. As the other category of shape optimization, deterministic optimization, just as the name implies, has definite search directions, thereby causing much reduction in time cost and correspondingly enhancing design efficiency [12], [13], [14], [15]. Thereinto, most of the deterministic optimization is gradient-based.

As one of gradient-based methods, adjoint method can obtain gradient information in the entire design domain through one iteration and thus is considered to be highly efficient. On this account, adjoint method that was originally used for optimization of hydrodynamics and mechanical structures has been introduced into EM field for shape optimization of EM components in recent years [16], [17], [18]. Specifically, the adjoint method and EM numerical methods, such as finite-element method and finite difference time domain method, are jointly used for shape optimization of EM components [19], [20]. It is noted that this combined approach needs to use the intermediate variables, i.e., system matrix and detail information of mesh generation provided by simulation programs. As is known, Maxwell equations' solving can be realized by customized simulation programs or commercial software packages. However, to the best of our knowledge, the current commercial EM software is unable to offer intermediate variables. It means that the combined

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Shengwei Ji, Liuge Du, Xiao Xu, and Jia Zhao are with the School of Information Science and Engineering, Shandong University, Qingdao 266237, China (e-mail: lgdu@sdu.edu.cn).

Siteng HuYan is with the Key Laboratory of Laser and Infrared System, Shandong University, Qingdao 266237, China.

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approach is not easily compatible with commercial software package, which makes it inconvenient and unfriendly [21], [22], [23].

Based on the above drawbacks, adjoint EM optimization independent on numerical methods is relatively more valuable. To enable commercial EM software packages to be used in conjunction with adjoint method, great efforts have been made. However, the developed approaches presented in the relevant articles [24], [25], [26] can only handle a single or several design DoF, such as moving a certain metal structural plane by optimization. Despite these drawbacks, the method is still more efficient than traditional methods mentioned above and it has been preliminarily implemented in commercial software [27], [28], [29]. There are some advanced strategies for dielectric optical device design with high DoF and some are integrated into optimization packages of commercial solvers due to the ease of obtaining the derivative of permittivity. For instance, Lumopt optimization package developed by Berkeley group is combined with commercial EM software Lumerical, and Stanford group has developed an automated photonic inverse design framework named Stanford Photonic INverse design Software (SPINS). Unfortunately, these integrated approaches developed for the field of optical waveguide devices are only applicable for optimizing dielectric topology or shape structures and powerless to deal with metal-based microwave components [15], [30], [31], [32].

In this article, we present a shape optimization approach for the inverse design of metal-based microwave components. It is independent of any specific numerical method, serving as a pure postprocessing procedure compatible with various commercial Maxwell solvers such as HFSS, CST Microwave Studio, and FEKO. This approach calculates shape derivatives at all the points in space, enabling high DoF in optimizing intricate structures while necessitating only one simulation per iteration.

The remainder of this article is organized as follows. Section II provides a brief description of the developed approach in this work and a detailed derivation of the related formulas. Section III shows two examples of designing microwave components adopting the proposed approach in this work. The validity of the design is demonstrated by a good agreement between the designed performance and the measured results. Finally, a conclusion is given in Section IV.

II. PRESENTATION OF THE PROPOSED APPROACH IN THIS WORK

In general, the inverse design of EM devices is to optimize the shape of the initial structures provided using a certain method, so that the EM performance of the final structures with optimized shape can meet the design objectives.

As is known, commercial EM software packages are incapable of providing system matrix and detail information of mesh generation, which is considerably indispensable for shape optimization of complex cases in the combined approaches jointly using adjoint method and customized simulation programs. Another way, as mentioned in the introduction, integrating adjoint method into EM software packages in [24], [25], [26] can only handle dimension optimization

with low DoF. This is because they only have dimension derivatives for structural surfaces and have not derived shape derivatives for each structural point. Hence, the proposed approach in this work needs another means to realize shape optimization in the absence of system matrix and detail information of mesh generation. And then the only means is to find a solution based on adjoint method itself. The certain method, i.e., the developed approach in this work, integrates adjoint method into commercial EM software packages while having the powerful capability of optimizing complex cases with high DoF, which is the most striking feature of this approach.

In this part, we elaborate adjoint method used in our developed approach and the key procedures involved in Section II-A. And then the derivation of shape derivative that is used for shape optimization is focused on in Section II-B. Finally, level set geometric representation that is applied to parameters' updating is introduced in Section II-C.

A. Principle of Adjoint Method for Electromagnetic Problems

At the beginning of this section, three reasonable assumptions are made for simplicity. First, in electromagnetic fields, changes in the electric field are accompanied by changes in the magnetic field. Therefore, the wave equation for the electric field is chosen for the derivation, and the electric field is selected as intermediate variable in the derivation process. Second, the metal considered in this work is perfect electrical conductor (PEC). Finally, the dielectric involved in this work is isotropic.

EM optimization design, by its very nature, is to find the optimal material distribution or geometry parameters in design region so that the maximum or minimum output can be reached at some point for a given input. In mathematical term, the EM design problem with general objective function f can be expressed as

$$\text{Minimax } f[\mathbf{E}(p_i)] \quad (1)$$

$$\text{Subject to } (\nabla \times \nabla \times - \omega^2 \epsilon \mu) \mathbf{E} = -j\omega \mu \mathbf{J} \quad (2)$$

where (1) is a composite function that maps the relationship between the objective function f and the design parameter $p_i (i = 1, \dots, n)$, $\mathbf{E}$ is the electric field, ω is the angular frequency, ϵ is the permittivity, μ is the permeability, and $\mathbf{J}$ is the current density.

As mentioned in the introduction, adjoint method is very efficient for solving the problem of EM optimization. In this method, adjoint simulation is the key and featuring step, which can combine forward simulation to find the optimal parameters in the design region. The operation principle is explained as follows. Since the frequency-domain wave equation is an implicit function of parameter p_i , according to the chain rule with the Wirtinger derivatives, the gradient of f can be expressed as

$$\frac{df}{dp_i} = \frac{\partial f}{\partial \mathbf{E}} \frac{d\mathbf{E}}{dp_i} + \frac{\partial f}{\partial \mathbf{E}^*} \frac{d\mathbf{E}^*}{dp_i} \quad (3)$$

where $*$ means the conjugation of electric field, and Re represents the real part. Since f is a real-valued function,

the Wirtinger derivatives are complex conjugates of each other

$$\frac{\partial f}{\partial \mathbf{E}} = \left(\frac{\partial f}{\partial \mathbf{E}^*} \right)^* \quad (4)$$

Then (3) can be written as

$$\frac{df}{dp_i} = 2\text{Re} \left(\frac{\partial f}{\partial \mathbf{E}} \frac{d\mathbf{E}}{dp_i} \right). \quad (5)$$

From the wave function (2), the derivative of the electric field with respect to a single design variable p_i is

$$\frac{\partial(\nabla \times \nabla \times - \omega^2 \epsilon \mu)}{\partial p_i} \mathbf{E} + (\nabla \times \nabla \times - \omega^2 \epsilon \mu) \frac{\partial \mathbf{E}}{\partial p_i} = -j\omega \mu \frac{\partial \mathbf{J}}{\partial p_i}. \quad (6)$$

Transform the position of the term in (6) to obtain

$$\begin{aligned} \frac{\partial \mathbf{E}}{\partial p_i} &= (\nabla \times \nabla \times - \omega^2 \epsilon \mu)^{-1} \\ &\times \left[-j\omega \mu \frac{\partial \mathbf{J}}{\partial p_i} - \frac{\partial(\nabla \times \nabla \times - \omega^2 \epsilon \mu)}{\partial p_i} \mathbf{E} \right]. \end{aligned} \quad (7)$$

Note that the source term $-j\omega \mu \mathbf{J}$ in (2) is independent of the parameter p_i , and hence the gradient of f can be written as

$$\begin{aligned} \frac{df}{dp_i} &= 2\text{Re} \left[\frac{\partial f}{\partial \mathbf{E}} (\nabla \times \nabla \times - \omega^2 \epsilon \mu)^{-1} \right. \\ &\quad \left. - \frac{d(\nabla \times \nabla \times - \omega^2 \epsilon \mu)}{dp_i} \mathbf{E} \right]. \end{aligned} \quad (8)$$

The middle item in (8) has the relationship of

$$\frac{\partial f}{\partial \mathbf{E}} (\nabla \times \nabla \times - \omega^2 \epsilon \mu)^{-1} = (\nabla \times \nabla \times - \omega^2 \epsilon \mu)^{-T} \left(\frac{\partial f}{\partial \mathbf{E}} \right)^T \quad (9)$$

where the superscript T means transpose of matrices. The adjoint electric field $\mathbf{E}^{ad}$ generated by the adjoint process is defined as

$$\mathbf{E}^{ad} = (\nabla \times \nabla \times - \omega^2 \epsilon \mu)^{-T} \left(\frac{\partial f}{\partial \mathbf{E}} \right)^T. \quad (10)$$

Then we can get the forward process form expressed as

$$(\nabla \times \nabla \times - \omega^2 \epsilon \mu) \mathbf{E} = -j\omega \mu \mathbf{J} \quad (11)$$

that satisfies the following two curl equations:

$$\nabla \times \mathbf{E} = -j\omega \mu \mathbf{H} \quad (12)$$

$$\nabla \times \mathbf{H} = j\omega \epsilon \mathbf{E} \quad (13)$$

and adjoint process form written as

$$(\nabla \times \nabla \times - \omega^2 \epsilon \mu)^T \mathbf{E}^{ad} = \left(\frac{\partial f}{\partial \mathbf{E}} \right)^T \quad (14)$$

that satisfies the following two curl equations:

$$\nabla \times \mathbf{E}^{ad} = -j\omega \mu \mathbf{H}^{ad} \quad (15)$$

$$\nabla \times \mathbf{H}^{ad} = j\omega \epsilon \mathbf{E}^{ad} \quad (16)$$

where $\mathbf{H}^{ad}$ represents the adjoint magnetic field generated by the adjoint process.

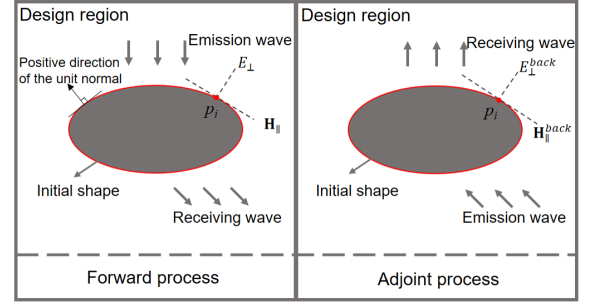


Fig. 1. Schematic of adjoint method for EM optimization problem and field solutions required for calculating parameter increments.

Equation (14) means that using $\partial f / \partial \mathbf{E}$ as a source, adjoint simulation generates an electric field $\mathbf{E}^{ad}$ based on the same structure as in the forward simulation. On the other hand, (11)–(16) indicate that the gradient in all the design regions can be calculated after both the forward simulation and adjoint simulation are performed. Fig. 1 illustrates the forward and adjoint process in adjoint method for EM optimization. It should be noted that this section only explains the feasibility of the adjoint method, and the adjoint process included cannot be directly calculated. The further processing and specific details will be discussed in Section II-B.

B. Shape Derivative in Microwave Structure Optimization

In this section, the derivation process of shape derivative for shape optimization of microwave devices will be presented.

In many researches concerning optimization of passive microwave devices, the aim is to obtain optimum transmission, reflection, absorption, radiation pattern, and so on of microwave structures. Since we will verify our method with specific devices later on, here we take an optimization objective in practical situations as an example for derivation. To distinguish from the general objective function f in Section II, figure-of-merit (FoM) is used to represent the specific optimization objective, which can be expressed by the Poynting vectors as

$$\text{FoM} = \frac{\iint (\mathbf{E} \times \mathbf{h}^{\text{back}}) \cdot d\mathbf{s}}{P_{\text{in}}} \quad (17)$$

$$P_{\text{in}} = \iint (\mathbf{E}^{\text{in}} \times \mathbf{h}) \cdot d\mathbf{s} \quad (18)$$

$$d\mathbf{s} = \mathbf{a}_n \cdot d\mathbf{s} \quad (19)$$

where $\mathbf{E}$ is the electric field resulting from the forward simulation, $\mathbf{E}^{\text{in}}$ is the incident field at the port, $\mathbf{h}^{\text{back}}$ and $\mathbf{h}$ are the magnetic port modal vectors, P_{in} is the incident wave, and $\mathbf{a}_n$ is the port unit normal vector.

The derivative of FoM with respect to the design parameter is

$$\begin{aligned} \frac{d\text{FoM}}{dp_i} &= 2\text{Re} \frac{\iint (\mathbf{h}^{\text{back}} \times \mathbf{a}_n) \cdot \frac{d\mathbf{E}}{dp_i} d\mathbf{s}}{P_{\text{in}}} \\ &= \text{Re} \left[\frac{-2j\omega}{P_{\text{in}}} \iint (\epsilon E_{\perp} \cdot E_{\perp}^{\text{back}} + \mu_0 \mathbf{H}_{\parallel} \cdot \mathbf{H}_{\parallel}^{\text{back}}) d\mathbf{s} \right] \end{aligned} \quad (20)$$

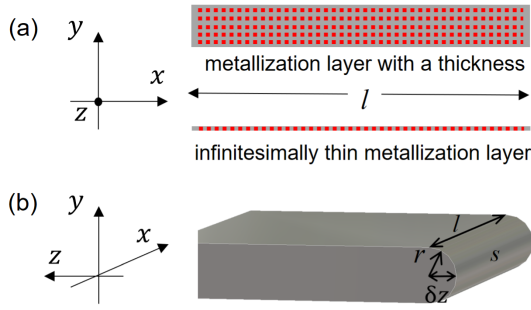


Fig. 2. (a) Schematic of metallization layer with a thickness and infinitesimally thin metallization layer. (b) Infinitesimally thin metallization layer with $r \rightarrow 0$ and $\delta z \rightarrow 0$.

where the subscript $\perp$ represents field component in the normal direction, and $\parallel$ represents field component in the tangential direction. It is noted that the actual optimization parameters in the shape derivative are the point displacements along the metal-boundary unit normal. The positive direction of the unit normal is demonstrated in Fig. 1.

Detailed derivation of (20) is reported in Appendix. Equation (20) is usually suitable for metallic size optimization, i.e., movement of metallic structural planes due to the surface integral in the last term of (20). In addition, it should be noted that sometimes it is necessary to deal with infinitesimally thin metallization layers, where the metallic plane may collapse into metallic lines, as shown in Fig. 2(a). Consider a metallization layer as shown in Fig. 2(b), when $r \rightarrow 0$ the metallization layer is infinitely thin, the origin surface integral transforms to a sensitivity expression involving integration over the surface of the half-cylinder [33]

$$\begin{aligned} \frac{d\text{FoM}}{dp_i} &= \text{Re} \left[\frac{-2j\omega}{P_{\text{in}}} \int_0^l \int_0^\pi (\epsilon E_z \cdot E_z^{\text{back}} + \mu_0 H_x \cdot H_x^{\text{back}}) r ds dl \right] \end{aligned} \quad (21)$$

where the subscript z represents field component in the z -direction, x represents field component in the x -direction, and l represents the length of the edge. The sampling at the edge of metallization layer is infeasible due to the singular. However, sampling can be performed along a line parallel to the respective edge and close to it inside the metal. Then (21) can be derived as

$$\begin{aligned} \frac{d\text{FoM}}{dp_i} &= \text{Re} \left[\frac{-j\omega\delta z\pi}{P_{\text{in}}} \int_0^l (\epsilon E_z \cdot E_z^{\text{back}} + \mu_0 H_x \cdot H_x^{\text{back}}) dl \right]. \end{aligned} \quad (22)$$

It can be seen that (22) corresponding to infinitesimally thin metallization layers and (20) for metallization layers with thickness only differ in terms of constant terms. However, the constant terms are not critical in practical applications. Therefore, as long as the correct field solutions can be provided, the gradient calculation of the adjoint method is not much different for infinitesimally thin metallization layers and metallization layers with a certain thickness.

To apply (20) to shape optimization of microwave structures, further derivation is required to optimize each point on the metallic boundary. The derivation of (20) in the context of perturbation theory for roughness losses is detailed in [34], [35]. Note the similarity of (20) to the variational equation of ordinary calculus

$$\delta\text{FoM} = \sum_i \frac{\partial\text{FoM}}{\partial p_i} \delta p_i \quad (23)$$

for a finite number of variables, a typical gradient-based optimization algorithm would operate by setting

$$\delta p_i = \frac{\partial\text{FoM}}{\partial p_i} \quad (24)$$

ensuring that to first order the objective function increased every iteration. Exactly the same reasoning leads in our case to the choice of

$$\delta p_i = \text{Re} \left[\frac{-2j\omega}{P_{\text{in}}} (\epsilon E_{\perp,i} \cdot E_{\perp,i}^{\text{back}} + \mu_0 \mathbf{H}_{\parallel,i} \cdot \mathbf{H}_{\parallel,i}^{\text{back}}) \right] \quad (25)$$

for all p_i on the boundary surface. Thus, right-hand side of (25) can be considered the infinite dimensional shape derivative as illustrated in Fig. 1, and we have a procedure for updating the shape with only one simulation.

C. Level Set Geometric Representation

For two-phase material microwave devices, the permittivity and conductivity are known everywhere once the boundary is specified. The level set methods provide a natural, flexible representation for arbitrarily shaped boundaries in a larger design space [36], [37]. For simplicity and clarity, we will take a practical example (i.e., the example of the coupler shown in Section III-A) to illustrate the process.

The level set function is expressed as $\phi[\mathbf{p}(t), t]$, where $\mathbf{p}(t)$ is the set of points along the boundary. The boundary can be deformed continuously through the artificial time parameter t and $t = 0$ represents the initial boundary. Within the boundary set is metal and outside the boundary set is air. Thus, the boundary shape of the structure can be represented by a zero level set

$$\phi[\mathbf{p}(t), t] = 0. \quad (26)$$

Taking the time derivative of (26) yields

$$\frac{\partial\phi}{\partial t} + \frac{d\mathbf{p}}{dt} \cdot \nabla\phi = 0. \quad (27)$$

In general, the term $d\mathbf{p}/dt$ is a vector which is referred to as the velocity, representing the speed at which a point along the boundary is moving. Since only movement normal to the boundary is relevant, movement along the boundary is simply a reparametrization. Then (27) can be rewritten as a Hamilton–Jacobi partial differential equation

$$\frac{\partial\phi}{\partial t} + V|\nabla\phi| = 0 \quad (28)$$

where V is a scalar velocity field in the normal direction and $|\nabla\phi|$ is the magnitude of the gradient of ϕ .

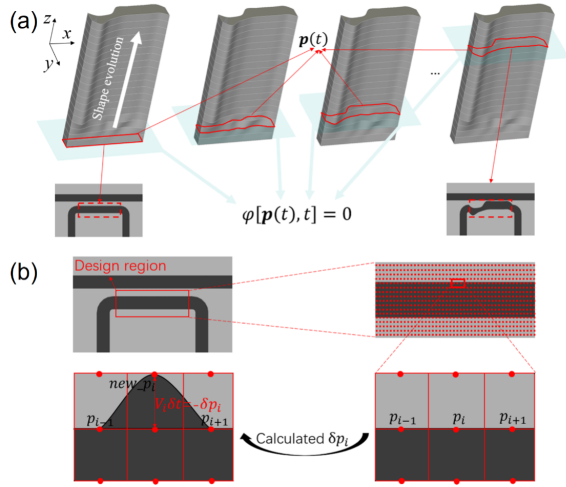


Fig. 3. (a) Level set method for parameters' updating in adjoint method. (b) Schematic of the relationship between δp_i calculated by adjoint method and V_i in the level set method.

A certain increment of a parameter δp_i which governs the deformation of the level set function can be expressed as

$$\delta p_i = \frac{\delta \phi_i}{|\nabla \phi_i|} = -\frac{dp_i}{dt} \delta t = -V_i \delta t. \quad (29)$$

Combining (22), (25), and (29) yields

$$\begin{aligned} \delta \text{FoM} \\ = \delta t \cdot \text{Re} \left[\frac{-2j\omega}{P_{\text{in}}} \iint V (\epsilon E_{\perp} \cdot E_{\perp}^{\text{back}} + \mu_0 \mathbf{H}_{\parallel} \cdot \mathbf{H}_{\parallel}^{\text{back}}) ds \right]. \end{aligned} \quad (30)$$

The variation in a parameter is

$$p_i^{\text{new}} = p_i - \delta p_i = p_i - \frac{\delta \phi_i}{|\nabla \phi|} = p_i + V_i \delta t \quad (31)$$

where p_i^{new} is the updated parameter and an updated structure can be obtained after calculating each p_i^{new} . The shape evolution of coupler's design region with the level set method is shown in Fig. 3(a). As can be seen, with the change in time t , the boundary $p(t)$ also changes. The deformation calculated by (25) is related to the velocity of boundary movement as demonstrated in (31) and Fig. 3(b).

The optimization process for microwave components is depicted in Fig. 4. Initially, the process commences with parametric modeling, where a parameter set is used to represent the device under optimization. Subsequently, simulation is conducted using commercial simulation software, with field monitors set at the desired frequencies prior to simulation. Upon completion of the simulation, the device's performance is assessed against the requirements, and if necessary, the iterative process is initiated.

The iterative process involves an adjoint simulation, wherein the position of the excitation source is adjusted to obtain the backward field value as illustrated in Fig. 2. Using the field solutions obtained from both forward and adjoint simulations, along with the provided equations, the shape derivative is calculated. With the derivative of the objective function calculated, changes in geometry are introduced proportionally to the derivative, using the gradient descent method. The shape

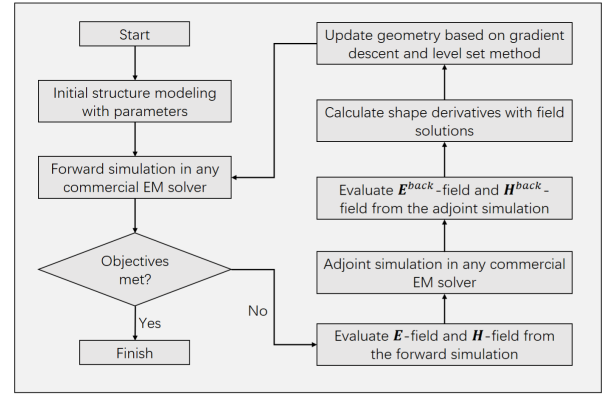


Fig. 4. Flowchart of the developed optimization approach in this work.

is updated using the level set method. The shape derivative calculated in the previous step as shown in Fig. 4 is used as the velocity field in the level set algorithm, resulting in a new shape for further simulation. Following the iterative process described above, the device structure that meets the design objectives is eventually obtained through repeated iterations.

III. VALIDATION AND DISCUSSION

To verify the validity of our developed approach, two types of microwave components including a directional coupler and an unequal Wilkinson power divider are designed according to the process in Fig. 4, fabricated and experimentally measured with respect to EM performance. The simulation software we use during the optimization process is CST Studio Suite and other EM software can also be used. Time-domain solver is chosen to simulate. The boundary conditions of the two devices on the x -axis, y -axis, and z -min are set as magnetic boundary, and the z -max direction is set as electric boundary. We have chosen the frequency point to be 5 GHz, so it is necessary to add electric field monitor and magnetic field monitor at 5 GHz both in the forward and adjoint simulations. The metallic layers of the two devices used to verify our method are not infinitely thin but have a thickness of 35 μm both in simulations and prototypes. Except for the simulations, all other optimization processes are completed using Python and embedded VB code. All the above processes are performed on a computer with a 16-GB RAM and an Intel Core i5-11400F running at 2.6 GHz.

A. High-Directivity Directional Coupler

The initial structure of the directional coupler is shown in Fig. 5, in which the coupling region within the red dashed box is the design region. Port 1 is specified as the input port, while ports 2, 3, and 4 are the through port, coupling port, and isolation port, respectively. The performance of directional couplers is usually evaluated by two metrics: coupling level (CL), which refers to the output at the coupling port, and directivity, which represents the difference between the coupling port and the isolation port. When CL increases, directivity also improves. Therefore, the optimization objective is set as CL and the objective function FoM_{CL} is

$$\text{maximize } \text{FoM}_{\text{CL}} = S_{31}. \quad (32)$$

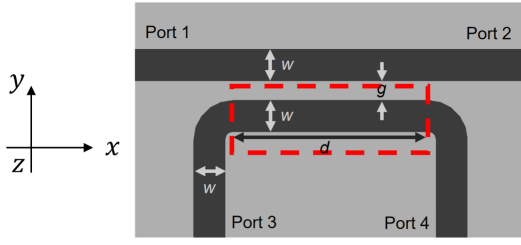


Fig. 5. Initial structure of the coupler.

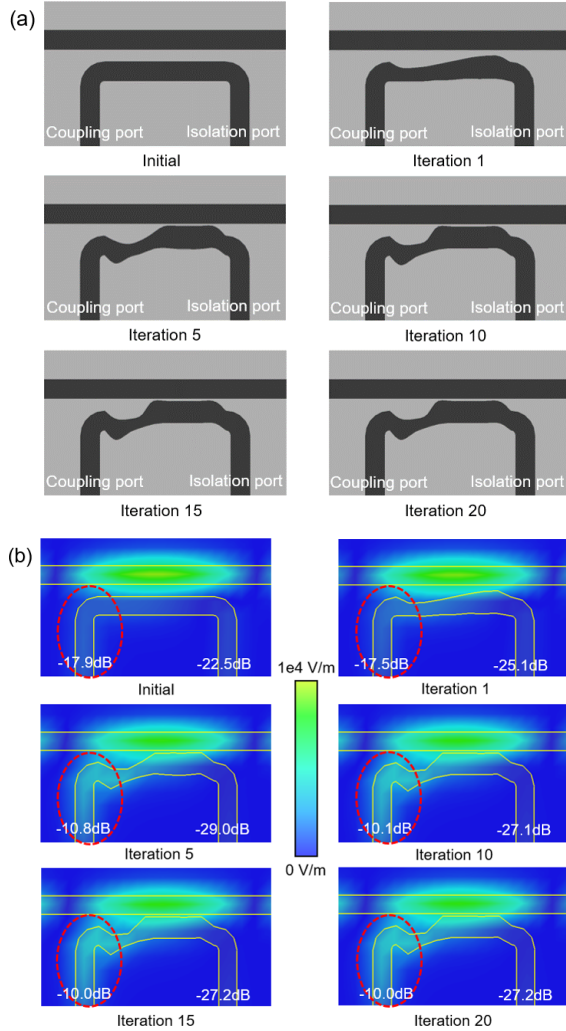


Fig. 6. Iterative optimization process of the coupler from the initial structure to the structure after 29 iterations. (a) Shape evolution. (b) Electric field distribution and the output of the coupling port and isolation port.

Given the actual situation, we have imposed limitations on the optimization area, which means that the upper edge of the coupling arm cannot exceed the through waveguide above, and a minimum spacing of 0.1 mm has been set. The initial geometry parameters are as follows, $w = 1.6$ mm, $d = 12$ mm, and $g = 1$ mm.

The shape evolution and the corresponding E -field distribution at 5 GHz during the optimization process are presented in Fig. 6. There is a visible shape change in the coupling region between the initial structure and the optimized structure.

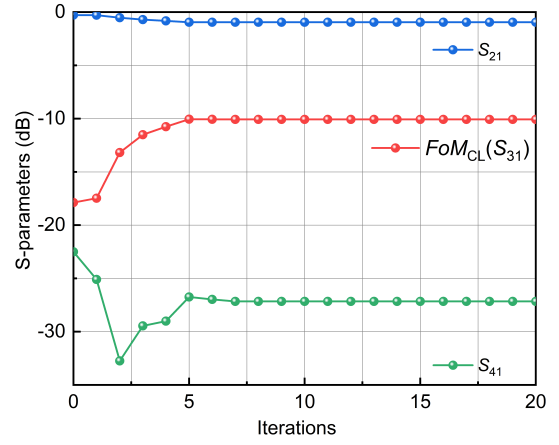


Fig. 7. S -parameters' evolution of coupler.

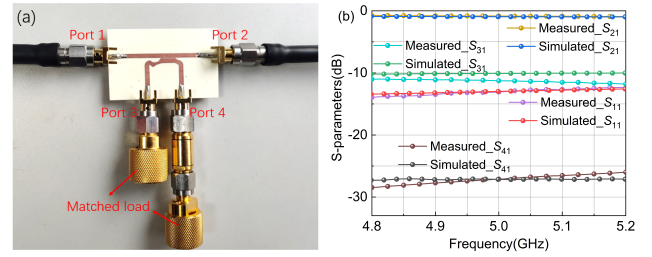


Fig. 8. (a) Prototype of the optimized high-directivity directional coupler. (b) Measured and simulated S -parameters of the optimized coupler.

More importantly, the E -field distribution on the coupled waveguide is obviously enhanced, while weakened on the isolation waveguide which means each iteration of the optimization process gets better. The S -parameters' evolution at 5 GHz is presented in Fig. 7, which shows the entire optimization process converges in less than ten iterations. The output of the coupling port increases from -17.9 to -10 dB and the directivity increased from 4.6 to 17.1 dB while maintaining low insertion loss below -1 dB. The total time of 20 iterations is about 55 min.

To verify the optimization results, a prototype of the optimized high-directivity directional coupler was fabricated using printed circuit board technology, in which 35- μ m-thick copper pattern was deposited on 30-mil-thick Rogers4350B substrate as shown in Fig. 8(a). The conductivity of copper is 5.8×10^7 S/m. The permittivity and loss tangent of Rogers4350B are 3.66 and 0.0037, respectively. The prototype is measured with Agilent vector network analyzer N5242A. The simulated and measured results of the fabricated coupler are plotted in Fig. 8(b). As can be observed, the four measured S -parameter curves are in accordance with the simulated results.

B. Unequal Wilkinson Power Divider

The second type of microwave structure used to verify our approach is a Wilkinson power divider as shown in Fig. 9. As illustrated by the red dotted box, the design region consists of a $\lambda/2$ long input waveguide and two $\lambda/4$ long splitter waveguides. The initial linewidth is $w = 2$ mm. The substrate is 30-mil-thick Rogers3003 in which permittivity and

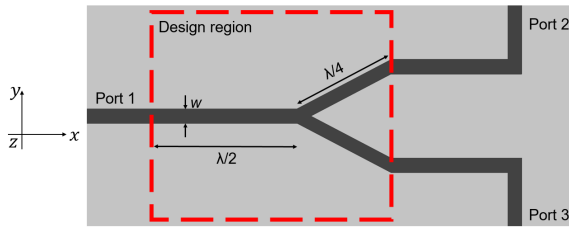


Fig. 9. Initial structure of the unequal Wilkinson power divider.

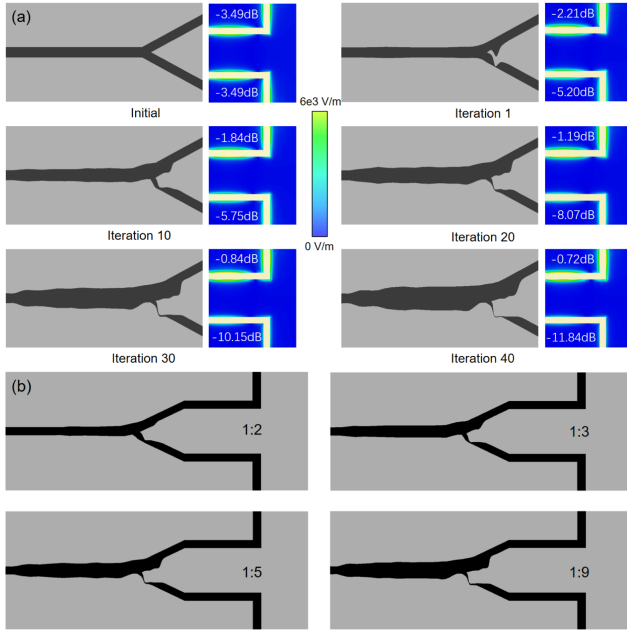


Fig. 10. Shape evolution of the unequal Wilkinson power divider during the optimization process. (a) Shape evolution and the corresponding field distribution from the initial structure to the structure after 40 iterations. (b) Shape structure corresponding to the power ratio of 2, 3, 5, and 9 in the optimization process.

loss tangent are 3 and 0.001, respectively. In the case of constant power input in power dividers, when the output of port 2 increases, the output of port 3 decreases accordingly. Therefore, the optimization objective of the unequal Wilkinson power divider is set to be the transmission coefficient (TC) of port 2 to realize the change in power division ratio. Therefore, the objective function FoM_{TC} is

$$\text{maximize } FoM_{TC} = S_{21}. \quad (33)$$

The shape evolution and the corresponding E -field distribution during the optimization process are presented in Fig. 10(a). Similarly, there is an apparent change between the initial structure and the final structure. The electric field distribution on port 2 shows an obvious enhancement while weakened on port 3. We enumerate the structural shapes with power distribution ratios of 2, 3, 5, and 9, as shown in Fig. 10(b). The deformation in the design area is larger for the structure with large power division ratio. It should be noted that to ensure the in-phase output of the two output ports, constraints on the structure description are added so that the microstrip lines are always connected. Fig. 11 illustrates that the output of port 2 increases during the iteration process as expected and the power divider can achieve an arbitrary

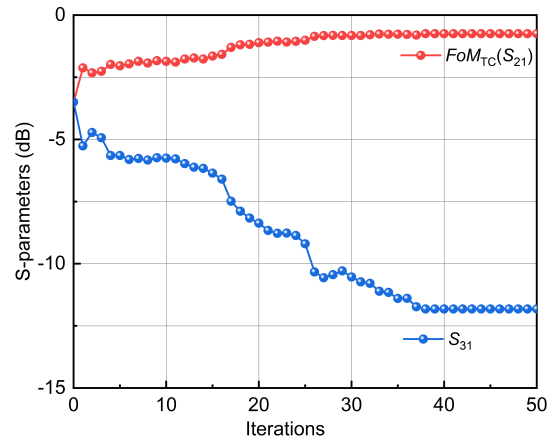


Fig. 11. S -parameters' evolution of the unequal Wilkinson power divider during optimization.

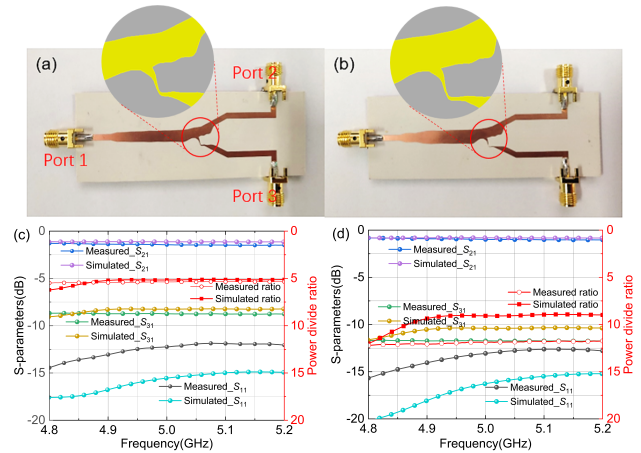


Fig. 12. Simulated results and measured results of the fabricated prototype. (a) Unequal power divider 1. (b) Unequal power divider 2. (c) Results of unequal power divider 1. (d) Results of unequal power divider 2.

TABLE I
COMPARISON WITH PUBLISHED UNEQUAL POWER DIVIDERS

	Center frequency	Bandwidth	Size(mm)	Ratio
Ref. [38]	1.5 GHz	2 GHz	30×47	2:1
Ref. [39]	5.8 GHz	0.2 GHz	33×46	11:1
Ref. [40]	6.5/6.5 GHz	7.5/7.5 GHz	31×27/31×27	2:1/3:1
Ref. [41]	2.4/2.4 GHz	2/2.4 GHz	75×40/95×45	4:1/8:1
Ref. [42]	3/3 GHz	4/3.6 GHz	80×50/82×59	2:1/1.2:1
Ref. [43]	3 GHz	2 GHz	76×48	8:1
This work	5 GHz	0.4 GHz	65×30/25×25 ¹	Arbitrary

¹ The data of 65×30 is the size we have expanded for the convenience of welding SMA ports and measurement, and the effective area size is 25×25.

ratio in a wide range. The total time of 50 iterations is about 3 h 24 min.

As shown in Fig. 12, two power dividers with power ratio of 1:5 and 1:9, respectively, were fabricated and measured. As seen from Fig. 12(a) and (b), it seems that the structure of the two power dividers is similar. Actually, as shown in the partial enlarged drawing in Fig. 12(b), the microstrip line leading to port 2 is wider while the microstrip line leading to port 3 is narrower compared with Fig. 12(a). The comparison between the simulated results and the measured results exhibits good consistency, thereby demonstrating the validity of the developed approach again.

TABLE II
COMPARISON WITH PUBLISHED EM OPTIMIZATION METHOD

		Optimization method ³	Optimize size or shape	Numerical method	Parameters	Iterations	Metal or dielectric
SO ¹	Ref. [44]	GWO	Size	Independent	6/8	100/100	Metal
	Ref. [45]	ML/SAPSO	Size	Independent	10	400 ⁴	Metal
	Ref. [46]	ONN	Size	Independent	5	100	Metal
GO ²	Ref. [21]	AM	Size	FDFD	4/5	9/7	Metal
	Ref. [25]	AM/NTF/TR	Size	Independent	5/4/5	6/7/6	Metal
	Ref. [26]	AM/NTF/TR	Size	Independent	5/6/6	9/6/6	Metal
	Ref. [47]	DSM/ADM	Size	Independent	10/9/7	14/14/20	Metal
	Ref. [48]	MRESM/TR	Size	Independent	15/10/8/12	10/9/10/8	Metal
	Ref. [19]	AM/SIMP/LSM	Shape	FEM	1109	240	Metal
	Ref. [49]	AM/SIMP/LSM	Shape	FEM	714	160	Metal
	Ref. [50]	AM/ADIM	Shape	BPM	480	55	Dielectric
	Ref. [51]	AM	Shape	BIM	19	40	Dielectric
	Ref. [52]	AM/BFGS	Shape	Independent	325	200	Dielectric
	Ref. [53]	AM/L-BFGS-B	Shape	Independent	>100	140/134/40	Dielectric
	This work	AM/GD/LSM	Shape	Independent	110/410	20/50	Metal

¹ SO represents stochastic optimization.

² GO represents gradient-based optimization.

³ The full name of the optimization methods are as follows. GWO: grey wolf optimization, ML: machine learning, SAPSO: surrogate-assisted particle swarm optimization, ONN: optimization neural network, AM: adjoint method, FDFD: finite-difference frequency-domain method, NTF: neuro-transfer function, TR: trust region, DSM: design specification management, ADM: automated decision-making, MRESM: multi-Resolution EM Simulations, SIMP: scalar isotropic material with penalization, LSM: level set method, FEM: finite-element method, AIDM: alternative direction implicit method, BPM: beam propagation method, BIM: boundary integral method, BFGS: Broyden–Fletcher–Goldfarb–Shanno, L-BFGS-B: limited-memory Broyden–Fletcher–Goldfarb–Shanno, GD: gradient descent.

⁴ 400 here refers to the simulations.

A comparison between our designed and optimized unequal power divider and the recently reported related works is shown in Table I. The most significant advantage of the power divider optimized by our method is its ability to easily achieve designs with arbitrary power ratios while also maintaining good performance in miniaturization. Furthermore, compared with the recently published work, our power divider structure does not rely on meticulously crafted designs based on experience. Instead, it requires only a simple initial structure, which is then automatically optimized.

C. Comparison With Other Methods

As outlined in the introduction, various optimization methods have been developed for EM component design, encompassing stochastic optimizations, and gradient-based optimizations. Here, we conduct a comparative analysis of our proposed method with several widely used techniques in recent years, as depicted in Table II. It is evident that while stochastic methods possess the advantage of global optimization, they are typically confined to size optimization with several parameters and generally require more iterations or simulations compared with gradient-based approaches. On the other hand, some gradient-based size optimization, although offering relatively less design freedom, necessitates fewer iterations, thereby achieving quicker convergence. In addition, shape optimization usually has more parameters compared with size optimization. In our approach, to achieve accurate boundaries, the finer the grid points are divided, the better the optimization effect will be. However, too many points will reduce the speed of calculations. Therefore, it is necessary to select an appropriate number of parameters. Taking microstrip devices in the frequency band of around 5 GHz as an example, based on our experience, a point spacing of around 0.1 to 0.5 mm is generally acceptable.

Due to efficient global gradient computation capability of the adjoint method, it is often integrated with numerical

methods, using information such as system matrices or mesh subdivision in numerical methods to achieve complex shape optimization, as evidenced in [18], [19], and [49]. Adjoint shape optimization independent of numerical methods is a valuable issue and has been explored in the field of optical devices, as discussed in [14], [15], [52], and [53].

In summary, our aim is to develop optimization methods that provide greater design flexibility, require fewer iterations, are user-friendly, and are applicable to microwave components. The method presented in this article has demonstrated favorable outcomes in these aspects.

IV. CONCLUSION

An adjoint-based shape optimization approach for the inverse design of microwave components has been presented. The proposed approach integrates adjoint method into commercial EM software, and gradient calculation and parameters' updating are the two key procedures. Gradient calculation is realized using rigorous analytical solution of shape derivative, while parameters' updating is achieved by geometric shape parameterization using the level set method.

The above-mentioned techniques endow the developed approach a few advantages. First, this approach only needs field solutions and is independent of any specific numerical method, serving as a pure postprocessing procedure compatible with various commercial Maxwell solvers. Second, this approach can achieve shape optimization with arbitrarily numerous DoF, while the reported approaches integrating adjoint method into commercial EM packages can only deal with several variables. Finally, the approach is highly efficient so that it only takes dozens or even a few iterations to accomplish optimization compared with other methods that usually require over a hundred iterations.

To validate our approach, we have applied it to optimize a directional coupler and an unequal power divider, followed by experimental measurements. The measured results agree well

with the optimized ones, thus demonstrating the validity and robustness of our developed approach.

APPENDIX

PROOF OF THE DERIVATIVE OF THE OBJECTIVE FUNCTION

The derivative of FoM with respect to the design variables p_i is

$$\frac{d\text{FoM}}{dp_i} = 2\text{Re} \frac{\iint (\mathbf{h}^{\text{back}} \times \mathbf{a}_n) \cdot \frac{d\mathbf{E}}{dp_i} ds}{P_{\text{in}}}. \quad (34)$$

The differentiation of two forward curl equations (12) and (13) with respect to parameter p_i produces

$$\nabla \times \frac{d\mathbf{E}}{dp_i} = -j\omega \left(\frac{d\mu}{dp_i} \mathbf{H} + \mu \frac{d\mathbf{H}}{dp_i} \right) \quad (35)$$

$$\nabla \times \frac{d\mathbf{H}}{dp_i} = j\omega \left(\frac{d\epsilon}{dp_i} \mathbf{E} + \epsilon \frac{d\mathbf{E}}{dp_i} \right) \quad (36)$$

and take the dot product of (35) with $\mathbf{H}^{\text{ad}}$, of (36) with $\mathbf{E}^{\text{ad}}$

$$\nabla \times \frac{d\mathbf{E}}{dp_i} \cdot \mathbf{H}^{\text{ad}} = -j\omega \left(\frac{d\mu}{dp_i} \mathbf{H} + \mu \frac{d\mathbf{H}}{dp_i} \right) \cdot \mathbf{H}^{\text{ad}} \quad (37)$$

$$\nabla \times \frac{d\mathbf{H}}{dp_i} \cdot \mathbf{E}^{\text{ad}} = j\omega \left(\frac{d\epsilon}{dp_i} \mathbf{E} + \epsilon \frac{d\mathbf{E}}{dp_i} \right) \cdot \mathbf{E}^{\text{ad}} \quad (38)$$

while take the dot product of (15) with $d\mathbf{H}/dp_i$, and of (16) with $d\mathbf{E}/dp_i$

$$\nabla \times \mathbf{E}^{\text{ad}} \cdot \frac{d\mathbf{H}}{dp_i} = -j\omega \mu \mathbf{H}^{\text{ad}} \cdot \frac{d\mathbf{H}}{dp_i} \quad (39)$$

$$\nabla \times \mathbf{H}^{\text{ad}} \cdot \frac{d\mathbf{E}}{dp_i} = j\omega \epsilon \mathbf{E}^{\text{ad}} \cdot \frac{d\mathbf{E}}{dp_i}. \quad (40)$$

Add the four equations (37)–(40) and use the vector identities

$$(\nabla \times \mathbf{A}) \cdot \mathbf{B} - (\nabla \times \mathbf{B}) \cdot \mathbf{A} = \nabla \cdot (\mathbf{A} \times \mathbf{B}) \quad (41)$$

to yield

$$\begin{aligned} \nabla \cdot \left(\mathbf{E}^{\text{ad}} \times \frac{d\mathbf{H}}{dp_i} + \mathbf{H}^{\text{ad}} \times \frac{d\mathbf{E}}{dp_i} \right) \\ = j\omega \left[\left(\frac{d\mu}{dp_i} \mathbf{H} \right) \cdot \mathbf{H}^{\text{ad}} - \left(\frac{d\epsilon}{dp_i} \mathbf{E} \right) \cdot \mathbf{E}^{\text{ad}} \right]. \end{aligned} \quad (42)$$

Integrating (42) and using divergence theorem to yield

$$\begin{aligned} \iint \left(\mathbf{E}^{\text{ad}} \times \frac{d\mathbf{H}}{dp_i} + \mathbf{H}^{\text{ad}} \times \frac{d\mathbf{E}}{dp_i} \right) ds \\ = j\omega \iiint \left[\left(\frac{d\mu}{dp_i} \mathbf{H} \right) \cdot \mathbf{H}^{\text{ad}} - \left(\frac{d\epsilon}{dp_i} \mathbf{E} \right) \cdot \mathbf{E}^{\text{ad}} \right] dv. \end{aligned} \quad (43)$$

By requiring the accompanying field to satisfy the same boundary conditions as the original field, the surface integration term on the left side of (43) can be correlated with the derivative of FoM [21]

$$\iint \left(\mathbf{E}^{\text{ad}} \times \frac{d\mathbf{H}}{dp_i} + \mathbf{H}^{\text{ad}} \times \frac{d\mathbf{E}}{dp_i} \right) ds = \frac{d\text{FoM}}{dp_i}. \quad (44)$$

The relationship between $\mathbf{E}^{\text{ad}}$ and $\mathbf{E}^{\text{back}}$ is as follows [21], [23], [54]:

$$\mathbf{E}^{\text{ad}} = -(2P_{\text{in}} \cdot P_{\text{in}}^{\text{back}})^{-1} \mathbf{E}^{\text{back}} \quad (45)$$

where $P_{\text{in}}^{\text{back}}$ is the power of incident wave in backward simulation.

Equation (45) indicates a linear relationship between the adjoint field and the backward field. Therefore, the right side of (44) can be made equal to the derivative of FoM of (34)

$$\begin{aligned} \frac{d\text{FoM}}{dp_i} &= 4\text{Re} \left\{ j\omega P_{\text{in}}^{\text{back}} \iiint \left[\left(\frac{d\mu}{dp_i} \mathbf{H} \right) \cdot \mathbf{H}^{\text{ad}} \right. \right. \\ &\quad \left. \left. - \left(\frac{d\epsilon}{dp_i} \mathbf{E} \right) \cdot \mathbf{E}^{\text{ad}} \right] dv \right\} \\ &= \text{Re} \left\{ \frac{-2j\omega}{P_{\text{in}}} \right. \\ &\quad \left. \times \iiint \left[\left(\frac{d\mu}{dp_i} \mathbf{H} \right) \cdot \mathbf{H}^{\text{back}} - \left(\frac{d\epsilon}{dp_i} \mathbf{E} \right) \cdot \mathbf{E}^{\text{back}} \right] dv \right\}. \end{aligned} \quad (46)$$

Equation (46) can be used in dielectric-based optical devices but not metal-based microwave devices; the derivative of permeability and permittivity have to be further derivation. Since the metal and dielectric are not magnetic

$$\frac{d\mu}{dp_i} = 0. \quad (47)$$

The relationship between permittivity and conductivity is

$$\epsilon = \epsilon_0 - \frac{j\sigma}{\omega} \quad (48)$$

then the derivative of the permittivity with respect to the parameter p_i is

$$\frac{d\epsilon}{dp_i} = \left[\epsilon_0(1 - \epsilon_r) + \frac{\sigma}{j\omega} \right]. \quad (49)$$

Substitute (49) into (46)

$$\begin{aligned} \frac{d\text{FoM}}{dp_i} &= \text{Re} \left[\frac{2j\omega}{P_{\text{in}}} \iint \epsilon_0(1 - \epsilon_r) \mathbf{E} \cdot \mathbf{E}^{\text{back}} ds \right] \\ &\quad + \text{Re} \left(\frac{2j\omega}{P_{\text{in}}} \iint \frac{\sigma}{j\omega} \mathbf{E} \cdot \mathbf{E}^{\text{back}} ds \right). \end{aligned} \quad (50)$$

From the Lorentz reciprocity

$$\begin{aligned} \nabla(\mathbf{E}_1 \times \mathbf{H}_2) &= \nabla(\mathbf{E}_2 \times \mathbf{H}_1) \\ &= -j\omega(\epsilon \mathbf{E}_1 \cdot \mathbf{E}_2 + \mu \mathbf{H}_1 \cdot \mathbf{H}_2). \end{aligned} \quad (51)$$

Using the expressions for the field behavior near PEC edges [23]

$$\iint (\epsilon \mathbf{E}_1 \cdot \mathbf{E}_2 + \mu \mathbf{H}_1 \cdot \mathbf{H}_2) ds = 0. \quad (52)$$

Substitute (48) into (52)

$$\iint \left[\left(\epsilon_0 - \frac{j\sigma}{\omega} \right) \mathbf{E}_1 \cdot \mathbf{E}_2 + \mu \mathbf{H}_1 \cdot \mathbf{H}_2 \right] ds = 0. \quad (53)$$

Simplify (53) and note that the tangential electric field and normal magnetic field are zero based on the above assumed PEC situation

$$\begin{aligned} \iint \frac{\sigma \mathbf{E} \cdot \mathbf{E}^{\text{back}}}{j\omega} ds \\ = - \iint (\epsilon_0 \mathbf{E}_{\perp} \cdot \mathbf{E}_{\perp}^{\text{back}} + \mu_0 \mathbf{H}_{\parallel} \cdot \mathbf{H}_{\parallel}^{\text{back}}) ds \end{aligned} \quad (54)$$

where $E_{\perp}$ means the normal component of electric field, and $\mathbf{H}_{\parallel}$ means the tangential component of magnetic field.

Note that the left side of (54) is part of (50). Substitute the right side of (54) into (50)

$$\frac{d\text{FoM}}{dp_i} = \text{Re} \left[\frac{-2j\omega}{P_{\text{in}}} \iint (\epsilon E_{\perp} \cdot E_{\perp}^{\text{back}} + \mu_0 \mathbf{H}_{\parallel} \cdot \mathbf{H}_{\parallel}^{\text{back}}) ds \right]. \quad (55)$$

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Shengwei Ji received B.S. degree from Qingdao University, Qingdao, China, in 2018, and the M.S. degree from Hefei University of Technology, Hefei, China, in 2021. He is currently pursuing the Ph.D. degree at Shandong University, Qingdao.

His research interests include electromagnetic inverse problem, microwave structure optimization, and metasurface design.

Siteng HuYan received the B.S. and M.S. degrees from Shandong University, Qingdao, China, in 2020 and 2023, respectively, where he is currently pursuing the Ph.D. degree.

His research interests include electromagnetic inverse problem, antenna test, and space electromagnetic field analysis.

Liuge Du received the B.S., M.S., and Ph.D. degrees in radio physics from Shandong University, Qingdao, China, in 2005, 2008, and 2011, respectively.

Currently, he is working as a Professor with the School of Information Science and Engineering, Shandong University. His research interests include theory and method of computational electromagnetics and simulation of semiconductor devices. He is also engaged in research on modeling and design of silicon photonics.

Xiao Xu received the B.S. degree in electronic science and technology from the University of Electronic Science and Technology of China, Chengdu, China, in 2014, and the Ph.D. degree in electric science and technology from Shanghai Jiao Tong University, Shanghai, China, in 2019.

She is working as an Associate Professor with the School of Information Science and Engineering, Shandong University, Qingdao, China. Her research interests include silicon and polymer photonic devices and short-reach optical interconnects.

Jia Zhao received the B.S. and the Ph.D. degrees from Shandong University, Qingdao, China, in 2006 and 2011, respectively.

He is currently working as a Professor with the School of Information Science and Engineering, Shandong University. His research interests include nanophotonics and chip-scale optical interconnection.